

"It was 80% me, 20% AI":

Seeking Authenticity in Co-Writing with Large Language Models

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Given the rising proliferation and diversity of AI writing assistance tools, especially those powered by large language models (LLMs), both writers and readers may have concerns about the impact of these tools on the authenticity of writing work. We examine whether and how writers want to preserve their authentic voice when co-writing with AI tools and whether personalization of AI writing support could help achieve this goal. We conducted semi-structured interviews with 19 professional writers, during which they co-wrote with both personalized and non-personalized AI writing-support tools. We supplemented writers' perspectives with opinions from 30 avid readers about the written work co-produced with AI collected through an online survey. Our findings illuminate conceptions of authenticity in human-AI co-creation, which focus more on the process and experience of constructing creators' authentic selves. While writers reacted positively to personalized AI writing tools, they believed the form of personalization needs to target writers' growth and go beyond the phase of text production. Overall, readers' responses showed less concern about human-AI co-writing. Readers could not distinguish AI-assisted work, personalized or not, from writers' solo-written work and showed positive attitudes toward writers experimenting with new technology for creative writing.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**; **Interactive systems and tools**.

1 INTRODUCTION

From text suggestion [40] and summarization [10] to style transformation [58], metaphor generation [36], and information synthesis [20], burgeoning applications of artificial intelligence (AI) for text production seem to be rapidly reshaping writing experiences and practices, especially with the recent high-profile releases of large language models (LLMs). Consequently, there are also concerns that vast transformations of the writer economy are likely underway [15, 37, 47, 53]. Within such a climate, seeking and preserving *authenticity*—as a cornerstone for all forms of creation—in writing content co-created with AI is likely to become an increasingly complicated yet critical matter for writers.

Indeed, existing literature has pointed to the importance of understanding authenticity for several reasons: From writers' perspectives, authenticity often determines the value of their work, which co-writing with AI might potentially threaten [18]. Moreover, writing serves as the medium for writers to connect with their audiences, and authentic expression contributes to the soundness of such bonds [7, 38, 52]. A deeper understanding of authenticity also facilitates discussions around ownership of work [13] and relevant practices such as declaring authorship, regulating copyright, detecting plagiarism, and commissioning writers' work. Recent work on AI use for writing has begun to explore relevant constructs, such as ownership, authorship, and agency [11, 13, 39, 49], but a more comprehensive understanding of

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views surrounding authenticity in human-AI co-creation remains elusive. Though public discussions reveal growing concerns from writers about the impact of AI on their work and profession [41], it remains unclear whether and how they would like to preserve elements of authenticity in writing.

Meanwhile, personalized AI applications—including *personalized AI writing assistance*—are becoming more common and readily accessible [9, 14, 27, 65]. This is believed to be especially promising with LLMs, which can be prompted or fine-tuned to generate a more specific form or style of text, allowing people with various degrees of AI/ML expertise to experiment with personalizing or steering text generation. For instance, a user could try to personalize AI writing suggestions to simulate their own writing style by specifying the characteristics of their desired style in the prompt or by providing a few of their own writing samples (i.e., in-context learning [3]). But can such personalization be sufficient to help preserve writers’ authentic voices in writing? While some recent research suggests personalized AI might add little to writers’ perceived ownership of their co-created writing work with AI [13], the potentials of and concerns about personalizing AI writing suggestions to support authenticity remain largely under-explored.

In this work, we take a closer look at authenticity in writing from both writers’ and readers’ perspectives. We focus on what writers seek for authenticity as new practices of co-writing with AI emerge and whether personalization could support their goals. Furthermore, as personalized AI tools become readily available, we seek to understand the possible impact of personalization on writers’ ability to express and preserve their authentic voices in writing. Specifically, we ask:

- RQ1: How do writers and readers conceptualize authenticity in the context of human-AI co-writing?
- RQ2: Based on their conceptions of authenticity, do writers want to preserve their authentic voices in writing, and if so, how?
- RQ3: Can personalized AI writing assistance support authenticity and help preserve writers’ authentic voices (if desired) in writing, and if so, how?

We examined these questions first through semi-structured interviews with 19 professional writers across various literature genres. During the interviews, writer participants reflected on their conceptions of authenticity in writing *and* co-writing with AI through situated experiences. Specifically, they engaged in writing with both generic and personalized AI writing assistance powered by a state-of-the-art large language model (GPT-4). We then complemented writers’ perspectives with those from avid readers through an online survey (N = 30), which allows us to gauge audiences’ responses to writers co-writing with AI and their perception of the authenticity of such work.

Our findings provide new insights into how writers and readers perceive co-writing with AI writing support. To begin with, writer-centered conceptions of authenticity focus more heavily on the internal experiences of writers and extend beyond the constructs of authenticity from prior literature (i.e., authenticity as source, category, and value of creators’ work) [21, 23, 51]. Furthermore, the use of AI raises several broader questions for creative writing: Can writers still be regarded as the sole sources creating the content when AI writing tools are used? Can the resulting work still compellingly capture and speak for the writers’ life experiences and the human stories informing the work? As writers reflected on these questions and practiced co-writing with AI during our study, they saw possible influences of AI-assisted writing on authenticity. While co-writing with AI did not fundamentally change their definitions (and thus understanding) of authenticity in creative work, they saw the need for and took various approaches to preserving their authentic voices in writing. This suggests an opportunity for the design of AI writing assistance tools to play an important role in supporting this endeavor. Finally, in contrast to writers’ concerns, in our study, readers expressed great interest in reading AI-assisted writing and were curious about how AI’s contributions might come into play.

Our work makes three key contributions: (1) We deepen our theoretical understanding of authenticity in the context of human-AI co-creation by surfacing and identifying writer-centered definitions of authenticity. These definitions incorporate aspects of authenticity that have not been accounted for in existing theories. (2) Our study offers design implications that underscore the need for co-writing tools, beyond their uses as productivity and creativity aids, to preserve authenticity in writing, ranging from supporting writers’ motivations and needs to addressing current pain points. (3) Finally, we also draw broader implications for the design of personalized AI tools, as they have been adopted in more and more creative domains. We discuss whether individual creators’ voices are amplified or lost when their own data is leveraged to create personalized AI tools.

2 BACKGROUND & RELATED WORK

2.1 Frameworks of Authenticity in Human Creative Work

The humanistic literature has long acknowledged authenticity as the core of human creative work [21, 42, 51]. Though there is a longstanding history of proposing theoretical frameworks for authenticity, researchers have not yet formed a consensus on the definition of authenticity. Still, they have often proposed three key themes to help define and conceptualize authenticity: *Category*, *Source*, and *Value* [2, 4, 8, 21, 23, 42, 43, 51, 62, 64]. First, the ***Category*** theme concerns whether a piece of work matches one’s existing beliefs about its associated category. The scope of a category can vary, ranging from a particular style or school of work (e.g., Bauhaus-style architecture) to a certain era (e.g., a Renaissance painting). Second, the idea of authentic ***Source*** concerns whether one can trace a piece of work to a specific source (i.e., a person, a place, an event, or any type of origin). This explains the importance of crediting writers and labeling the origins of work. In particular, when work from certain individuals is truly one-of-a-kind—such as the highly recognizable work of Picasso—audiences can easily identify them as the source of content. In such cases, the concepts of *Category* and *Source* become more blended and interchangeable. Finally, the ***Value*** theme is about whether there is consistency between a creator’s internal states and their external expression. This focuses on whether a writer’s perspectives, opinions, and values are consistent with what they expressed in their work.

In our study, we explore whether writers’ perceptions of authenticity in writing align with these concepts of authenticity for broader creative work and whether specific concerns apply to writing. Based on writer-centered definitions of authenticity we uncover, we further examine whether co-writing with AI is considered authentic and whether emerging AI technologies change writers’ views about the essence of authenticity.

2.2 The Impact of AI Use on Authenticity in Writing

2.2.1 Writers’ growing concerns regarding AI writing assistance. The increasing popularity of using AI for creative tasks has motivated recent work to investigate the possible impact of AI on several aspects of creators’ work, including credit, authorship, ownership, control, and agency [11, 13, 25, 35], many of which are closely related to authenticity. Recent studies, workshops, panels, and other forms of discussion [41] have thus far revealed mixed opinions from research communities, creators, and the general public toward these topics. Here, we summarize a few emergent themes:

Writers remain hesitant to declare AI co-authorship publicly. Recent work on perceived authorship of AI-generated text reveals a complex *AI Ghostwriter Effect* [13]. Through comparing personalized and pseudo-personalized AI tools, the findings suggest that though writers did not perceive complete ownership over AI-generated text, they were also reluctant to publicly declare AI co-authorship. This reluctance could be related to writers’ concerns about negative responses from their readers [18]. But more importantly, writers feel that authorship should go hand in hand